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# Strong invariance principles for diffusions, Markov chains and their perturbations

V. Konakov[1,2], D. Kucher[1], E. Mammen[3]

[1]HSE University, Moscow
[2]Lomonosov Moscow State University
[3]Heidelberg University

# Setting

A diffusion in $\mathbb{R}^d$ and its perturbation ($W$ is a $d$-dimensional Brownian motion):

$$dX_t = b(t, X_t)\,dt + \sigma(t, X_t)\,dW_t, \qquad dX^\epsilon_t = b_\epsilon(t, X^\epsilon_t)\,dt + \sigma_\epsilon(t, X^\epsilon_t)\,dW_t.$$

A Markov chain: Euler scheme with step $h = T/n$ and centred errors $\xi^n$,

$$Y_{t_{i+1}} = Y_{t_i} + b(t_i, Y_{t_i})\,h + \sigma(t_i, Y_{t_i})\sqrt{h}\,\xi^n_{i+1}, \qquad t_i = ih,$$

and $Y^\epsilon$: the same dynamics with $b_\epsilon, \sigma_\epsilon$ (linearly interpolated between grid points).

**Goal:** construct two processes on one probability space such that $\sup_{[0,T]}$ of their distance is small with explicit probability, for an arbitrary horizon $T$.

# Assumptions

**A1** boundedness $|b| + |\sigma| \le K$ (also for $b_\epsilon, \sigma_\epsilon$)

**A2** ellipticity $\lambda^{-1}|\xi|^2 \le \langle a\,\xi, \xi\rangle \le \lambda|\xi|^2, \quad a = \sigma\sigma^*$

**A3** Hölder continuity $|b(t,x) - b(s,y)| \le B\big(|x-y| + |t-s|^{\beta}\big)$

$|\sigma(t,x) - \sigma(s,y)| \le A\big(|x-y|^{\gamma_2} + |t-s|^{\alpha_2}\big)$

**A4** error densities $|D_z^\nu q^n_{t,x}(z)| \le C_q\, Q_S(z), \quad |\nu| \le 5, \quad S > 2d+6$

$|q^n_{t,x}(z) - q^n_{t,y}(z)| \le C_q\sqrt{h}\,|x-y|\, Q_S(z)$

**Markov case only;** $q^n_{t,x}$: density of the error given state $x$ at time $t$

Here $Q_S(z) = c_S(1+|z|)^{-S}$, $\int Q_S = 1$; all constants do not depend on $n$, $T$; $\gamma_2, \alpha_2, \beta \in (0,1]$.

# Theorem 1. Perturbed diffusions

No Markov chain here: both processes are diffusions driven by the same $W$; there are no errors $\xi^n$. Assume A1–A3. For all $m \in \mathbb{N}$, $\alpha \in (\frac{1}{2}, 1)$, $r > d$ there is a coupling of $X$ and $X^\epsilon$ with

$$\mathbb{P}\Big( \sup_{s \le T} |X_s - X_s^\epsilon| > 2m^{\alpha-1} + \frac{2KT}{m} \Big) \le 4dm \, \exp\Big( - \frac{m^{2\alpha-1}}{2dK^2T} \Big) + C_1(r,T)\, m\, \Delta_{\epsilon,\gamma,r},$$

where $\Delta_{\epsilon,\gamma,r} = \sup_t |\sigma - \sigma_\epsilon|_\gamma + \sup_t \|b - b_\epsilon\|_{L^r}$ (Hölder norm, $\gamma = \gamma_2$).

Growth in $T$: $C_1(\infty, T) \approx \exp\big(cT^{1/\gamma_2}\big)$, $c$ independent of $T$. Here $r = \infty$: $b - b_\epsilon$ is measured in the sup-norm; for $r < \infty$ (integral $L^r$-norm) the power depends on $r$ and is $\ge 1/\gamma_2$.

# Structure of the proof of Theorem 1

Fix the macro-grid $\tau_k = kT/m$, $k = 0, \dots, m$, with cells $[\tau_k, \tau_{k+1}]$ of length $T/m$.

1. **Transition densities:** $p$ and $p_\epsilon$ are close over one cell.
2. **Grid distributions:** the laws of $(X_{\tau_k})_{k \le m}$ and $(X^\epsilon_{\tau_k})_{k \le m}$ are close in total variation.
3. **Maximal coupling:** $X_{\tau_k} = X^\epsilon_{\tau_k}$ for all $k = 0, \dots, m$, up to probability $C_1\, m\, \Delta_{\epsilon,\gamma,r}$.
4. **From the grid to paths:** complete both paths between the grid points.
5. **Fluctuations:** inside each cell both processes move by at most $m^{\alpha-1} + KT/m$.

Steps 1–4 give the term $C_1\, m\, \Delta_{\epsilon,\gamma,r}$, Step 5 gives $4dm \exp\big(-\frac{m^{2\alpha-1}}{2dK^2T}\big)$.

# Step 1. Transition densities

Parametrix: freeze the coefficients at the terminal point $y$ and expand

$$p = \sum_{N \geq 0} \widetilde{p} \otimes H^N, \qquad p_\epsilon = \sum_{N \geq 0} \widetilde{p}_\epsilon \otimes H_\epsilon^N,$$

where $\widetilde{p}$ is Gaussian, $H$ is the parametrix kernel and $H^N$ its $N$-fold convolution. Comparing the series term by term (Konakov–Kozhina–Menozzi 2017) and summing them gives

$$|p - p_\epsilon|(s,t,x,y) \leq C_1(r,T)\, \Delta_{\epsilon,\gamma,r}\; p_c(t-s, y-x),$$

with a Gaussian density $p_c$. The sum over $N$ is a Mittag-Leffler series: this is where $C_1(r,T)$ comes from. Integrating in $y$:

$$\int |p - p_\epsilon|(\tau_{k-1}, \tau_k, x, y)\, dy \leq C_1(r,T)\, \Delta_{\epsilon,\gamma,r} \qquad \text{for all } x \in \mathbb{R}^d,\ k = 1, 2, \ldots, m.$$

# Step 2. Distributions on the grid

Let $\mathbf{p}, \mathbf{q}$ be the densities of $(X_{\tau_1}, \dots, X_{\tau_m})$, $(X^\epsilon_{\tau_1}, \dots, X^\epsilon_{\tau_m})$, $X_0 = X^\epsilon_0 = x$. By the Markov property, with $p_k = p(\tau_{k-1}, \tau_k, z_{k-1}, z_k)$, $p^\epsilon_k = p_\epsilon(\tau_{k-1}, \tau_k, z_{k-1}, z_k)$,

$$\mathbf{p} - \mathbf{q} = \prod_{k=1}^{m} p_k - \prod_{k=1}^{m} p^\epsilon_k = \sum_{i=1}^{m} \Big( \prod_{k<i} p^\epsilon_k \Big) \, (p_i - p^\epsilon_i) \, \Big( \prod_{l>i} p_l \Big).$$

In the $i$-th term integrate $z_m, \dots, z_{i+1}$ (each $p_l$ is a density: integral $1$), then $z_i$ by Step 1, then $z_{i-1}, \dots, z_1$ (each $p^\epsilon_k$ is a density). Every term is at most $C_1 \Delta_{\epsilon,\gamma,r}$, so

$$\|\mathbf{p} - \mathbf{q}\|_{L^1} \le C_1(r, T)\, m\, \Delta_{\epsilon,\gamma,r}.$$

# Step 3. Maximal coupling

Put $\mathbf{h} = \min(\mathbf{p}, \mathbf{q})$ and $W = \int \mathbf{h} = 1 - \frac{1}{2}\|\mathbf{p} - \mathbf{q}\|_{L^1}$. Sample $V = (X_{\tau_1}, \dots, X_{\tau_m})$ with density $\mathbf{p}$, and

- with probability $\mathbf{h}(V)/\mathbf{p}(V)$: set $V^\epsilon = V$;
- with probability $1 - \mathbf{h}(V)/\mathbf{p}(V)$: sample $V^\epsilon$ from the residual density $\mathbf{r} = \dfrac{\mathbf{q} - \mathbf{h}}{1 - W}$, independently of $V$.

Then $V^\epsilon$ has density $\mathbf{q}$: for every set $A$, $\mathbb{P}(V^\epsilon \in A) = \int_A \mathbf{h} + (1 - W) \int_A \mathbf{r} = \int_A \mathbf{q}$. Moreover,

$$\mathbb{P}(V \neq V^\epsilon) = \int \Big(1 - \frac{\mathbf{h}}{\mathbf{p}}\Big)\mathbf{p} = 1 - W = \tfrac{1}{2}\|\mathbf{p} - \mathbf{q}\|_{L^1} \leq C_1(r, T)\, m\, \Delta_{\epsilon,\gamma,r}.$$

# Step 4. From the grid to paths

Set $X_{\tau_k} := V_k$, $X^\epsilon_{\tau_k} := V^\epsilon_k$. Given $(V, V^\epsilon)$, sample the path of each process from its conditional law given its grid values, independently for the two processes.

- Existence: regular conditional laws exist, since paths lie in the Polish space $C[0,T]$.
- Correct law: (law of the grid values) $\times$ (conditional law of the path) $=$ law of $X$ on $C[0,T]$.
- Markov property: the conditional law splits into independent bridges over the cells $[\tau_{k-1}, \tau_k]$.
- The grid values are not changed: $\mathbb{P}(A) \geq 1 - C_1(r,T)\, m\, \Delta_{\epsilon,\gamma,r}$ for $A = \{X_{\tau_k} = X^\epsilon_{\tau_k}\ \forall k\}$.

# Step 4. The coupling is kept

Let $B = \left\{ \sup_{s \le T} |X_s - X^\epsilon_s| > 2m^{\alpha-1} + \frac{2KT}{m} \right\}$. Then

$$\mathbb{P}(B) \le \mathbb{P}(A^c) + \mathbb{P}(A \cap B).$$

- On $A$ the processes coincide at the grid points, so $\mathbb{P}(A \cap B)$ is controlled by the fluctuations of $X$ and $X^\epsilon$ inside the cells.
- Fluctuations: the fluctuation bound depends only on the law of the path in $C[0,T]$, and this law is the same for the completed process and for $X$. Hence the bound of Step 5, proved for $X$, holds for the completed process.

# Step 5. Fluctuations in the cells

For $t \in [\tau_k, \tau_{k+1}]$: $X_t - X_{\tau_k} = \int_{\tau_k}^t b(s, X_s)\, ds + M_t$, $M_t = \int_{\tau_k}^t \sigma(s, X_s)\, dW_s$.

- Drift: $\big|\int b\, ds\big| \le K\,T/m$.
- Martingale: each component $M^i$ is a continuous local martingale with $\langle M^i\rangle_t \le K^2(t-\tau_k)$. The exponential inequality for continuous local martingales (Revuz–Yor, Ex. IV.3.16) gives $\mathbb{P}\big(\sup_t |M_t^i| \ge x\big) \le 2\exp\big(-\frac{x^2 m}{2K^2T}\big)$. Take $x = m^{\alpha-1}/\sqrt{d}$, all $d$ components and $m$ cells:

$$\mathbb{P}\Big(\max_k \sup_{t\in[\tau_k,\tau_{k+1}]} |X_t - X_{\tau_k}| > m^{\alpha-1} + \frac{KT}{m}\Big) \le 2dm\,\exp\Big(-\frac{m^{2\alpha-1}}{2dK^2T}\Big).$$

The same holds for $X^\epsilon$.

# Conclusion of the proof

With $A = \{X_{\tau_k} = X^\epsilon_{\tau_k} \text{ for all } k\}$ and $B = \big\{ \sup_{s \le T} |X_s - X^\epsilon_s| > 2m^{\alpha-1} + \frac{2KT}{m} \big\}$ as in Step 4,

$$\mathbb{P}(B) \le \mathbb{P}(A^c) + \mathbb{P}(A \cap B).$$

On $A$ the processes coincide at the grid points, so for $s \in [\tau_k, \tau_{k+1}]$

$$|X_s - X^\epsilon_s| \le |X_s - X_{\tau_k}| + |X^\epsilon_s - X^\epsilon_{\tau_k}|, \qquad \text{hence} \qquad A \cap B \subset F \cup F^\epsilon,$$

where $F, F^\epsilon$ are the fluctuation events of Step 5 for $X$ and $X^\epsilon$. Therefore

$$\mathbb{P}(B) \le \underbrace{C_1(r,T)\, m\, \Delta_{\epsilon,\gamma,r}}_{\mathbb{P}(A^c)} + \underbrace{2 \cdot 2dm\, \exp\Big( - \frac{m^{2\alpha-1}}{2dK^2T} \Big)}_{\mathbb{P}(F)+\mathbb{P}(F^\epsilon)}.$$

Choice of $m$: if $\Delta \to 0$, take $m \to \infty$ with $m\Delta \to 0$.

# Theorem 2. Degenerate diffusions

A similar theory can be developed for degenerate diffusions (again driven by $W$, no errors $\xi^n$).

For $d\widetilde{X}_t = b(\widetilde{X}_t, \widetilde{Y}_t)\,dt + \sigma(\widetilde{X}_t, \widetilde{Y}_t)\,dW_t$, $d\widetilde{Y}_t = \widetilde{X}_t\,dt$ and its perturbation with $b_\epsilon, \sigma_\epsilon$: assume A1, A2 and Hölder continuity of order $\gamma_3$ in $x$, $\gamma_3/3$ in $y$. For all $m$, $\alpha \in (\frac{1}{2}, 1)$, $r > 4d$,

$$\mathbb{P}\Big( \sup_{s \le T} |\widetilde{X}_s - \widetilde{X}^\epsilon_s| > 2m^{\alpha-1} + \frac{2KT}{m} \Big) \le 4dm\, \exp\Big( - \frac{m^{2\alpha-1}}{2dK^2T} \Big) + C_2(r,T)\, m\, \overline{\Delta}_{\epsilon,\gamma,r},$$

where $\overline{\Delta}_{\epsilon,\gamma,r}$ is defined as $\Delta_{\epsilon,\gamma,r}$, with the Hölder norm in the metric $|x - x'| + |y - y'|^{1/3}$, $\gamma = \gamma_3$.

Growth in $T$: $C_2(\infty, T) \approx \exp\big(cT^{1/\gamma_3}\big)$; for $r < \infty$ the power depends on $r$ and is $\ge 1/\gamma_3$.

# Degenerate case: the Kolmogorov density

For $d\widetilde{X} = dW$, $d\widetilde{Y} = \widetilde{X}\,dt$ $(\widetilde{X}_u, \widetilde{Y}_u)$ is Gaussian (Kolmogorov, 1934); with the covariance multiplied by $2/c$ its density is

$$p_{c,K}\big(u,(x,y),(x',y')\big) = \frac{c^d\,3^{d/2}}{(2\pi u^2)^d}\exp\Big(-c\Big(\frac{|x'-x|^2}{4u} + \frac{3\,|y'-y-\frac{(x+x')u}{2}|^2}{u^3}\Big)\Big).$$

Stability (Kozhina 2018, Thm. 4.3.1): under A1, A2, AD3, for $r \in (4d, \infty]$,

$$|p - p_\epsilon|\big(t_i, t_j, (x,y),(x',y')\big) \le C_2(r,T)\,\overline{\Delta}_{\epsilon,\gamma,r}\; p_{c,K}\big(t_j - t_i, (x,y),(x',y')\big).$$

# Markov chains: two regimes for the errors $\xi^n$

**Independent errors**

i.i.d. in time, smooth density with polynomial decay:

$$|D^\nu f_\xi^n(z)| \le C_\xi \, Q_M(z), \quad |\nu| \le 5$$

$\to$ **Theorem 3**: chain vs. perturbed chain

**Markov errors**

the law of the next error depends only on the current state:

$$\mathcal{L}\big(\xi_{i+1}^n \,\big|\, Y_{t_i}^\epsilon = y_i,\, Y_{t_{i-1}}^\epsilon, \dots\big) = q_{t_i, y_i}^n$$

$\to$ **Theorem 4**: chain vs. diffusion

Theorems 1–2 concern diffusions only: the noise is $W$, there are no errors $\xi^n$.

# Theorem 3. Chain vs. perturbed chain

We now come to another setting: comparison of a Markov chain with its perturbation, independent errors.

Assume A1–A3; the errors $\xi_k^n$ are independent with density $f_\xi^n$, $|D^\nu f_\xi^n| \le C_\xi Q_M$, $|\nu| \le 5$. For $m$ dividing $n$, $\alpha \in (\frac{1}{2}, 1)$ and every $R > 0$:

$$\mathbb{P}\Big( \sup_{s \le T} |Y_s - Y_s^\epsilon| \ge 2m^{\alpha-1} + \frac{2KT}{m} + \frac{2c\sqrt{Tn}}{m\, R^{M-d-1}} \Big)$$
$$\le 4dm \, \exp\Big( - \frac{m^{2\alpha-1}}{2dR^2T} \Big) + \frac{2c\,n}{R^{M-d}} + C_3(T)\, m\, \Delta_{\epsilon,\gamma,\infty},$$

where $c$ depends only on $C_\xi, M, d, K$. $R > 0$ is a free parameter: the coupling does not depend on $R$.

Growth in $T$: $C_3(T) \approx \exp\big(cT^{1/\gamma_2}\big)$ (here always $r = \infty$, the sup-norm).

# Theorem 4. Setting: Markov chain vs. diffusion

Finally we consider the comparison of a Markov chain with a diffusion, Markov errors.

The chain:

$$Y^{\epsilon}_{t_{i+1}} = Y^{\epsilon}_{t_i} + h\, b_\epsilon(t_i, Y^{\epsilon}_{t_i}) + \sqrt{h}\,\eta_{i+1}, \qquad \eta_{i+1} = \sigma_\epsilon(t_i, Y^{\epsilon}_{t_i})\,\xi^n_{i+1}.$$

- Noise: given the past, $\eta_{i+1}$ (the increment without the factor $\sqrt{h}$) has the density $q^{\epsilon,n}_{t_i,x}$ at $Y^{\epsilon}_{t_i} = x$, with mean $0$ and covariance $a^\epsilon(t_i, x)$.
- A1–A3 are imposed on the coefficients: $(b, \sigma)$ of the diffusion and $(b_\epsilon, a^\epsilon)$ of the chain.
- A4 is imposed on the noise densities $q^{\epsilon,n}_{t,x}$ (in place of $q^n_{t,x}$), uniformly in $n$.

# Theorem 4. Markov chain vs. diffusion

Under the assumptions of the previous slide (Markov errors), for $n \geq T$, $m$ dividing $n$, $\alpha \in (\frac{1}{2}, 1)$ and every $R > 0$:

$$\mathbb{P}\Big( \sup_{s \leq T} |X_s - Y_s^\epsilon| \geq 2m^{\alpha-1} + \frac{2KT}{m} + \frac{c\sqrt{Tn}}{m\, R^{S-d-1}} \Big)$$
$$\leq 2dm\Big( \exp\Big( -\frac{m^{2\alpha-1}}{2dK^2T} \Big) + \exp\Big( -\frac{m^{2\alpha-1}}{2dR^2T} \Big) \Big) + \frac{c\, n}{R^{S-d}} + C_4(T)\, m\, \widetilde{\Delta}^{(n,m)},$$

$$\widetilde{\Delta}^{(n,m)} = \ln\Big(\frac{en}{m}\Big)\Big(\frac{T}{m}\Big)^{\frac{\gamma_2 - 1}{2}} \Delta^{(n)} + \Big(\frac{m}{n}\Big)^{1/2},$$
$$\Delta^{(n)} = h^{\min(\frac{\gamma_2}{2}, \alpha_2, \beta)} + \sup |a - a^\epsilon| + \sup |b - b_\epsilon|.$$

Growth in $T$: $C_4(T) \approx \exp\big(c\, T^{2(S+1)/\gamma_2}\big)$.

# Proofs of Theorems 2–4: the same steps

- **Coupling on the grid** (Steps 1–4): the same argument, with the corresponding bound for the transition densities: degenerate diffusions (Theorem 2), two chains (Theorem 3), chain and diffusion (Theorem 4).
- **Fluctuations in the cells** (Step 5): for diffusions as before; for chains, large errors are handled separately and the Azuma–Hoeffding inequality replaces the martingale inequality.
- The new ingredient of Theorem 4 is the bound for the transition densities of the chain and the diffusion.

# Theorem 4, Step 1: transition densities

The new ingredient. For all grid points $t_i < t_j$ and $x, y \in \mathbb{R}^d$, with $\tau = t_j - t_i$,

$$|p - p_\epsilon|(t_i, t_j, x, y) \le C_4(T) \Big[ \ln\big(e(j-i)\big)\, \tau^{\frac{\gamma_2 - 1}{2}} \Delta^{(n)} + (j-i)^{-1/2} \Big] \, Q_{S-d-6}\Big(\frac{x-y}{\sqrt{\tau}}\Big),$$

where $Q_{S-d-6}(\cdot/\sqrt{\tau})$ is normalised to integral $1$.

- Proof: parametrix expansions $p = \sum_N \widetilde{p} \otimes H^N$, $p_\epsilon = \sum_N \widetilde{p}_\epsilon \otimes_\epsilon H^{\epsilon,N}$, compared term by term.
- With $j - i = n/m$ and integration in $y$:
  $\int |p - p_\epsilon|(\tau_{k-1}, \tau_k, x, y)\, dy \le C_4(T)\, \widetilde{\Delta}^{(n,m)}$.
- Extends Bitter–Konakov (2026) to an arbitrary horizon $T$, with explicit $C_4(T)$.

# Choice of parameters

Take $n = m^l$ and $R = n^\theta$ with

$$l > \max\Big(3,\ \frac{3-\gamma_2}{2\min(\gamma_2/2,\ \alpha_2,\ \beta)}\Big), \qquad \frac{1}{S-d} < \theta < \frac{2\alpha-1}{2l}.$$

If $\sup|a-a^\epsilon| + \sup|b-b_\epsilon| = o\big(m^{-(3-\gamma_2)/2}/\ln m\big)$, then with probability $\to 1$ as $m \to \infty$

$$\sup_{s\in[0,T]} |X_s - Y_s^\epsilon| \le \big(2 + o(1)\big)\, m^{\alpha-1}.$$

The condition $l > 3$ ensures $m\,(m/n)^{1/2} = m^{(3-l)/2} \to 0$ (the term $(m/n)^{1/2}$ in $m\,\widetilde{\Delta}^{(n,m)}$).

# Summary

- Strong approximations uniformly on $[0, T]$, for any horizon $T$, with explicit constants.
- Diffusions and degenerate diffusions (Theorems 1–2).
- Independent errors: chain vs. perturbed chain (Theorem 3).
- Markov errors: chain vs. diffusion (Theorem 4).
- Scheme: coupling on a grid (transition densities $\to$ total variation $\to$ maximal coupling), then fluctuations in the cells.

Thank you!